

The Three Structural Failures of DCF

Session 2 • Jensen Bias • McKean-Vlasov • LA-CAPM

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

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What we'll cover today

1

Recap: where DCF fails

Session 1 anchored the \$73M problem

2

Failure 1: Stochastic premium

Secondary market shows 10× moves vs. constant assumption

3

Failure 2: Jensen convexity bias

$E[f(X)] \neq f(E[X])$ when discount factor is nonlinear

4

Failure 3: McKean-Vlasov externality

Liquidity is collective, not individual

5

LA-CAPM

Connection to portfolio theory you already know

The three failures, now with detail

FAILURE 1

Stochastic Premium

Secondary market data shows premia move from 3% to 50%. Constant-assumption DCF cannot price what the market clears at. Resolution: Ornstein-Uhlenbeck process for liquidity state.

→ Sessions 3, 4

FAILURE 2

Jensen Convexity Bias

Discount factor $\exp(-rT)$ is convex in r . When r is stochastic, $E[\exp(-rT)] > \exp(-E[r]T)$. DCF substitutes $E[r]$, systematically undervaluing assets.

→ Sessions 11, 24

FAILURE 3

McKean-Vlasov Externality

Liquidity provision is a collective action problem. The marginal LP's exit affects all others' execution prices. Equilibrium requires general equilibrium framework.

→ Sessions 21, 24

Each failure has a closed-form correction. Together, they constitute GE-LAV.

Failure 1: A constant premium contradicts the data

Headline: secondary market discounts move 10× the magnitude of the assumed constant premium.

Empirical evidence (US PE secondary market):

Regime	Discount	Context
2007 peak	3–8%	Boom: premium near floor
2009 GFC trough	35–50%	Liquidity crisis: premium 10× normal
2020 COVID	15–25%	Brief shock; recovered in 9 months
2022 rate shock	15–20%	Sustained ~24 months
2024 (current)	12–18%	Still well above 3.5% constant

DCF assumes 3.5% constant. Data shows 5–50% range, mean-reverting.

Resolution: model the liquidity state as a stochastic process (OU). Premium = function of state.

Failure 2: Jensen convexity bias

Jensen's inequality applied to the discount factor:

$$E[e^{-rT}] > e^{-E[r] \cdot T}$$

What this means

When the discount rate r is stochastic and the discount factor $\exp(-rT)$ is convex in r , the expected value exceeds the value at the expected rate.

The asset is worth MORE than DCF says.

Why DCF gets it wrong

DCF substitutes $E[r]$ into the discount factor, computing $\exp(-E[r]T)$ — strictly less than $E[\exp(-rT)]$.

Calibrated magnitude: 0.8–3.6%/yr by asset class.

Jensen bias: a worked numerical example

Setup: 20-year asset, \$50M annual CF. Premium $\sim N(3.5\%, \sigma^2)$ with $\sigma = 0$ vs $\sigma = 1.5\%$.

Scenario	σ (premium vol)	Mean rate	DCF value	True value (Jensen)	Bias
Deterministic	$\sigma = 0\%$	7.5%	\$103.6M	\$103.6M	0%
Mildly volatile	$\sigma = 1.5\%$	7.5%	\$103.6M	~\$108M	~4%
Highly volatile (GFC)	$\sigma = 5.0\%$	7.5%	\$103.6M	~\$135M	~30%
State-dependent (GE-LAV)	Varies	Time-varying	\$103.6M	\$30.9M (depth)	Negative!

Key insight: Jensen says LAV > DCF (positive bias).

But GE-LAV < DCF in liquidity crises — because the McKean-Vlasov externality (Failure 3) dominates the Jensen correction (Failure 2).

Two opposing forces. GE-LAV captures both.

Failure 3: Liquidity is collectively determined

The McKean-Vlasov externality — a thought experiment

Individual perspective

One LP decides to sell a \$10M PE stake on the secondary market.

Buyers are deep, time-to-clear ~ 60 days, discount ~ 8%.

LP takes this as the equilibrium price for the asset.

→ DCF logic: *discount = function of asset's individual properties.*

Collective perspective

1,000 LPs each decide to sell a \$10M PE stake at the same time.

Total supply = \$10B. Secondary market depth = ~\$3B/quarter.

Discount blows out to 35–50%. Time-to-clear ~ 12 months.

→ *Discount = function of the collective state, not the asset.*

Each LP's decision affects every other LP's execution price. No individual internalizes this.

This is the McKean-Vlasov externality. It is irreducible to individual-asset DCF analysis.

LA-CAPM: connection to portfolio theory

Acharya & Pedersen (2005) — Liquidity-Adjusted CAPM

Standard CAPM:

$$E[r_i] - r_f = \beta_i \cdot (E[r_m] - r_f)$$

One source of risk: comovement with market return.

Liquidity-Adjusted CAPM:

$$E[r_i] - r_f = E[c_i] + \beta_i^{(1)}(E[r_m] - r_f) + \beta_i^{(2)} + \beta_i^{(3)} - \beta_i^{(4)}$$

β^1 : market beta

comovement with market return (standard)

β^2 : liquidity-liquidity

comovement of asset liquidity with market liquidity

β^3 : return-liquidity

asset return when market is illiquid

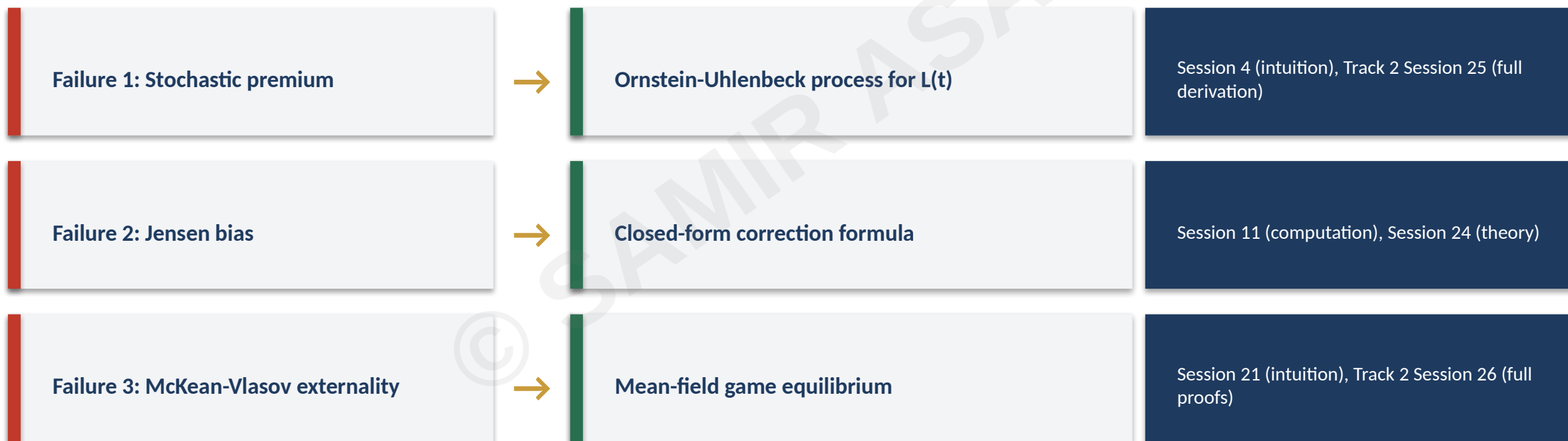
β^4 : liquidity-return

asset liquidity when market is down

GE-LAV generalizes LA-CAPM with: (1) general equilibrium pricing, (2) collective dynamics (McKean-Vlasov), (3) full term structure.

Three failures, three resolutions, three later sessions

Roadmap for the three corrections:



Session 2 summary

What we accomplished today

- 1 Three structural failures of DCF: stochastic premium, Jensen bias, McKean-Vlasov externality
- 2 Each failure has empirical support and a mathematical resolution
- 3 Jensen bias is positive ($LAV > DCF$), McKean-Vlasov is negative — GE-LAV captures both
- 4 LA-CAPM (Acharya-Pedersen) is the partial-equilibrium precursor; GE-LAV is its general-equilibrium generalization

Next session

Session 3: Liquidity Illusion · the secondary market evidence in depth · the Liquidity-Adjusted CAPM in practice